I design, build and run AI automation in production for a London accountancy firm.

A construction tax statement pipeline with reconciliation dashboards, an LLM assistant on WhatsApp and the web, and a bank statement tool for quarterly tax figures, in use at Trusted Figures.

Looking for an automation, integration or solutions engineering role in London, starting February 2027.

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01 / 04

CIS statement pipeline and dashboards

- Live: intake, matching, filing and client dashboards; statement requests in first use

41	client dashboards at launch, Aug 2026[1]
335 of 351	documents received since July with a final outcome 22 Jul to 13 Oct 2026, WhatsApp and email, Oct 2026[2]

Staff see, for every client, which contractor tax statements are still missing, without checking by hand.

02 / 04

AI chatbot on WhatsApp and the website, with a team dashboard

- Live; one firm in production

42 of 42	acceptance benchmark cases passed live AI model, Jul 2026[3]

Clients get answers on WhatsApp and the website; staff step in from one dashboard.

03 / 04

Making Tax Digital figures from bank statements

- In internal use for quarterly Making Tax Digital submissions

140 of 140	real statements self-verified 14,801 transactions, 0 parse failures, Jul 2026[4]
9	bank statement layouts read 11 UK bank brands, 10 tested on real statements, Aug 2026[5]

Turns a sole trader's bank statements into checked figures for their quarterly tax update.

04 / 04

Business automations in n8n

- Live

14	active workflows live n8n instance, Oct 2026[6]

Client messages, documents, registrations and filing deadlines are handled in the background; staff are told what needs them.

01 of 04

CIS statement pipeline and dashboards

Staff see, for every client, which contractor tax statements are still missing, without checking by hand.

Status	Stack	Role
Live: intake, matching, filing and client dashboards; statement requests in first use	Python, PostgreSQL, Self-hosted OCR, OpenAI, n8n	Designed and built

41
client dashboards
at launch, Aug 2026[1]

335 of 351
documents received since July with a final outcome
22 Jul to 13 Oct 2026, WhatsApp and email, Oct 2026[2]

What it does

Clients

Send statements by WhatsApp or email, as a photo, PDF or spreadsheet.

Get a WhatsApp request in English or Georgian listing the statements still missing.

Staff

Pull every client's expected deductions from QuickBooks after one attended two-factor login.

See expected against collected per client; click any month for figures, reasons and files.

Open any statement, renamed and filed by client, from the hub, PC or web.

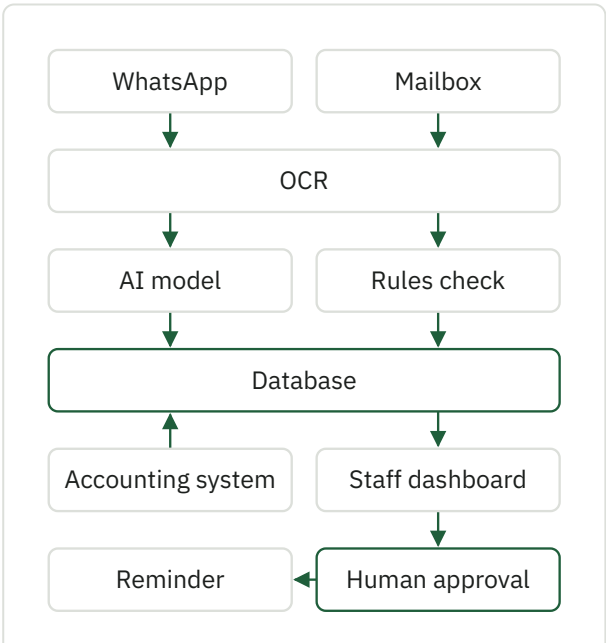
Correct any expected figure; an append-only ledger records who, when and why.

Problem

Construction subcontractors have tax deducted by every contractor they work for, and claiming it back needs a statement from each contractor for each month. The statements arrived scattered over WhatsApp and email and were reconciled once a year by hand, so discrepancies surfaced only at year end and delayed clients' repayment claims.

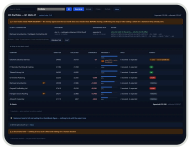
What I built

A pipeline built to run each quarter, with the first quarter complete. It starts from the accounting system's expected deductions, then reads every statement that arrives, matches it to its expected month and files it, holding anything unclear for a person. Staff track every client on one dashboard.



Under the hood

- Nothing is dropped: every document that arrives gets a recorded outcome, from matched to held for a person.





- A statement is filed only when client, contractor and month are all certain; the sender's identity is used only to break a tie.
- OCR runs on the firm's own machine, so no page image goes to a cloud OCR service. An LLM reads the extracted text and a rules check re-reads month and amount; where they disagree, the rules win on a known contractor layout, otherwise the document is held.
- Regression set: after each fix to the reader, **47**[7] real documents are read again and compared with a person's recorded answers.
- Scheduled checks run on live data: weekly invariants, a heartbeat per job and a contradiction check of every published page.
- Databases and stored files are backed up nightly, encrypted; a rehearsed restore in October matched every count.
- Python and PostgreSQL, money as exact decimals, **4,293**[8] automated tests; a snapshot test fails if a change alters a client's own page, which is built without firm-wide data.

Results

- **62 of 64**[9] Q1 2026-27 statements identified were matched to what each client owed.
- **17**[10] duplicate copies caught by file fingerprint and counted once.
- QuickBooks export: about 4 minutes of manual download per company before, now **under a minute**[11].

Decisions and trade-offs

- Hold rather than guess: more documents wait for a person, but no wrong match reaches a client's folder.
- Statement requests are attended: each client's expected list is approved and checked first, and a person confirms each send, one client at a time. It costs staff time, but a wrong or repeated message cannot reach a client.

02 of 04

AI chatbot on WhatsApp and the website, with a team dashboard

Clients get answers on WhatsApp and the website; staff step in from one dashboard.

Status	Stack	Role
Live; one firm in production	TypeScript, Supabase, n8n, WhatsApp Cloud API, OpenAI, Anthropic	Designed and built

42 of 42

acceptance benchmark cases passed

live AI model, Jul 2026[3]

What it does

Clients

Ask questions on WhatsApp or the website chat in 6[12] languages.

Ask what we need and get back the request we sent, never a guess.

Register for CIS (construction tax) or a taxpayer reference in website chat, uploading ID.

Send several files on WhatsApp and get one reply confirming how many arrived.

Staff

Take over any thread and reply under their own name; the bot goes silent.

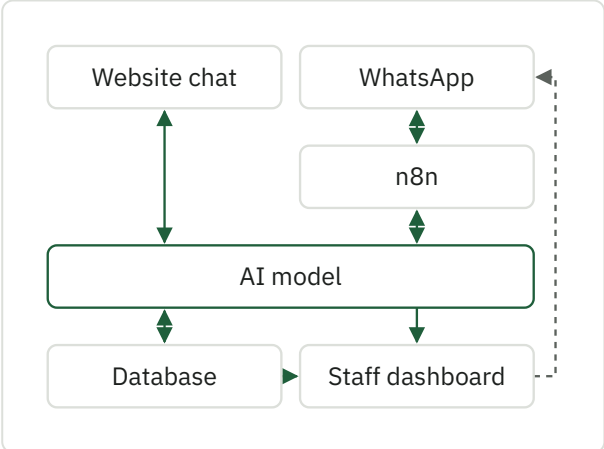
Send messages and files; after WhatsApp's 24-hour reply window, approved templates only.

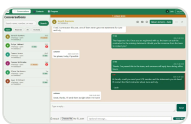
Admins

Check company filing deadlines, pull a client's files and set reminders by WhatsApp command.

Problem

Clients would rather ask a question or send a photo on WhatsApp than write an email, and the construction tax (CIS) statement pipeline needed a client-facing channel where clients already are. A growing office cannot answer every message quickly and consistently, and a wrong tax rate or another client's details in a reply is a real risk.









What I built

An LLM assistant that answers clients on WhatsApp and the firm's website chat. It is also the client-facing front end of the CIS statement pipeline: files sent to the WhatsApp number are queued for the pipeline, confirmed with one reply, and alerted to staff if one fails to save. Staff read every thread on a team dashboard and can take any of them over.

Under the hood

- The model is instructed to answer only from the firm's service material, and code checks every reply for prices, tax rates, formatting and confidentiality before it is sent.

- An evaluation set of **49**[13] scripted client conversations, 26 of which must pass on every run, goes through the bot's real reply route against the live model.
- The main reply call has a timeout, two retries and a circuit breaker, then fails over to a second AI provider.
- Spending caps per firm and per user, and a per-user rate limit, stop a fault or misuse from running up AI costs.
- The bot masks common UK identifiers in each new message and in stored memory lines.
- An hourly sync copies a short record per client from the CIS pipeline into the bot's own database, with no money fields; the bot never reads the CIS database.
- TypeScript on Supabase, with n8n carrying every WhatsApp message in and out; **1,689**[14] automated tests.

Decisions and trade-offs

- Redirect, never guess: tax-rate questions always get a GOV.UK link set in code, and a client can ask for a person at any point. The bot answers fewer questions on its own, but no client gets a guessed rate.
- Unattended messages are pre-approved templates only, each checked for duplicates, opt-outs, deletion requests and business hours, and a blocked send is logged with its reason. Reminders are less flexible, but no free text reaches a client without a person.
- Confidentiality is enforced after generation: a reply that names a company not recorded against the sender is replaced by a refusal. It can refuse a safe answer, and refuses in English only so far, but a reply naming another client's company does not go out.

Debugging story

A client sent eight photos at once and only some were stored. Each new file was first saved under the same placeholder name, which the database accepts only once, so photos arriving together were rejected. Each file now gets its own placeholder; a test reproduces the loss, and a live burst test after the fix stored **`8 of 8`**[15] photos, each paired to its message.

03 of 04

Making Tax Digital figures from bank statements

Turns a sole trader's bank statements into checked figures for their quarterly tax update.

Status	Stack	Role
In internal use for quarterly Making Tax Digital submissions	Python, PDF text parsing, Rules engine	Designed and built

140 of 140
real statements self-verified
14,801 transactions, 0 parse failures, Jul 2026[4]

9
bank statement layouts read
11 UK bank brands, 10 tested on real statements, Aug 2026[5]

What it does

Accountant

Load a client's PDF statements; any that fail the bank's own totals are refused.

Review every uncertain line in one list before it counts towards a figure.

Answer grouped questions on uncertain lines, largest amounts first, before saving a rule. built, not in daily use yet

Set rules for one client or a whole trade; client rules override trade rules.

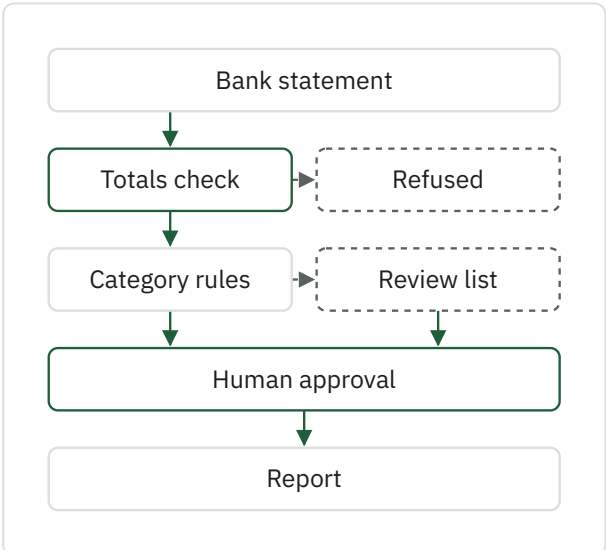
Get the self-employment return boxes (SA103), a review workbook and the filing spreadsheet.

Problem

Making Tax Digital asks sole traders for quarterly figures, and each quarter starts from bank statements whose every line must be read, classed as income or expense and totalled. By hand it takes a long time, and external tools cost too much for the work to pay. One misread or misclassified line also changes the figure that gets filed.

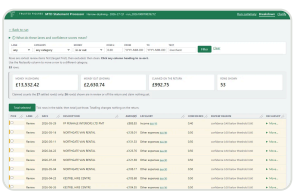
What I built

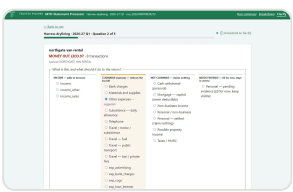
A local tool that parses each bank's PDF statements, verifies them against the bank's own totals and classifies every line with layered fixed rules. A person always reviews and files; the tool sends nothing to HMRC.



Under the hood

- A statement is trusted only when its lines re-add to the bank's printed totals and running balance to the penny; a balance break is shown at its row.
- No AI and no network calls in reading or sorting; an automated check blocks common network and AI libraries.





- Rules sit in three layers, client over trade over firm, and one client can switch off a trade rule without touching anyone else.
- Money is held as exact decimals, never floating point, and the tool never modifies a statement file.
- Python, with lint, strict type checks and **976**[16] automated tests.

Results

- A rule change meant for one client would have moved a figure on another client's filed return. A regression test pinning a sample client's return boxes to the penny caught it before it landed.
- After rules were reordered, cash taken out at a supermarket cashpoint was about to count as a claimable meal cost. An automated test caught it before any return was filed.

Decisions and trade-offs

- Fixed rules, no AI: the same statements and rules always give the same figures, and every category traces back to a rule. A new client means more questions for the accountant at first, but no figure moves without a reason.
- Read the text layer the PDF already holds, not a picture of the page. Scanned statements are refused for now, but every amount is taken as the bank printed it, and the totals check proves it.
- Local and single user: the page answers only on the same computer and has no login, so one person uses it today. A hosted version for more staff waits until it has proved itself on real quarters.

04 of 04

Business automations in n8n

Client messages, documents, registrations and filing deadlines are handled in the background; staff are told what needs them.

Status	Stack	Role
Live	n8n, WhatsApp Cloud API, Microsoft Graph, Companies House API, Supabase	Designed and built

14
active workflows
live n8n instance, Oct 2026[6]

What it does

Staff

Get WhatsApp and email reminders before Companies House filing deadlines, and when overdue.

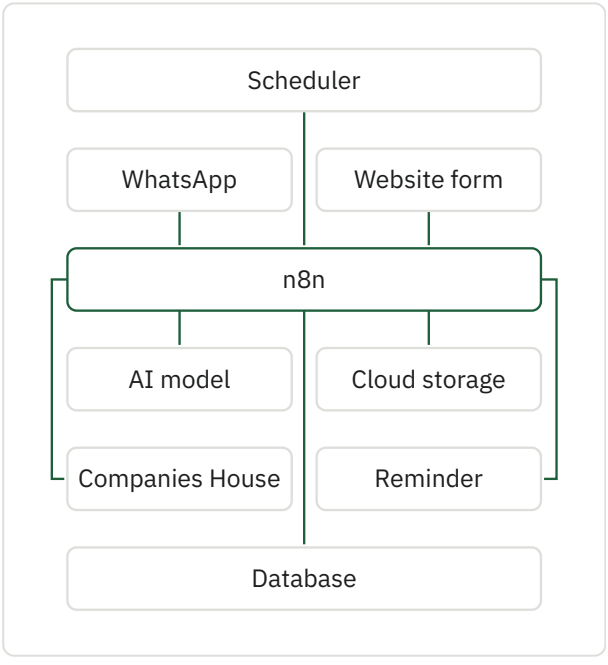
Work from a deadline tracker refreshed from Companies House every day.

Receive a WhatsApp alert with folder link and photo count per website registration.

Get an alert when a filing step fails, saying what to file by hand.

Problem

Client work arrives on WhatsApp, by email and through a website form, and every client company has Companies House filing deadlines. Done by hand it is slow, and a dropped step or a missed deadline goes unnoticed until it is late.



What I built

Background workflows in n8n. The **14**[17] active workflows, by job:

Messages in

2 workflows

WhatsApp text goes to the chatbot, files to the document pipeline; an older email route runs only when started by hand.

Messages out

2 workflows

Template messages leave through two guarded send workflows: client messages from the firm's apps, and document filing alerts to staff.

Document filing

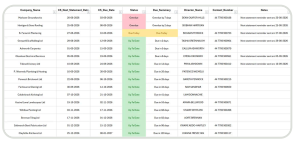
3 workflows

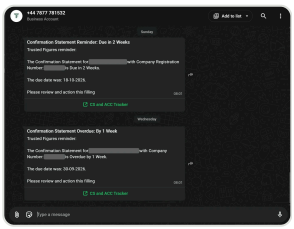
Each read document is filed in the right client folder in cloud storage; two helpers clear stale copies and empty folders.

Registrations

1 workflow

Each website registration becomes a client folder with its ID photos and a details file.





Compliance `5 workflows`

Two daily trackers check Companies House and remind staff, one daily sync feeds the team dashboard, two refresh rows on request.

Monitoring `1 workflow`

A 30-minute check emails the owner once when the chatbot's client records go stale, again as a reminder, and on recovery.

Also built, now switched off: the chatbot's first version, which searched FAQs and HMRC guidance and drafted answers with an AI model. It never answered a client.

Under the hood

- Every webhook the firm's own apps call checks a shared secret or token, and no credentials are stored in the workflow files.
- Reminder sends from the firm's apps reserve a one-time idempotency key before sending, so a duplicate is refused, and make up to 3 attempts.
- Incoming WhatsApp files are checked against allowed types and de-duplicated on the provider's message id before they are queued.
- Only approved templates go out and blank fields are refused; filing retries before alerting staff, and a folder is deleted only when it is empty.

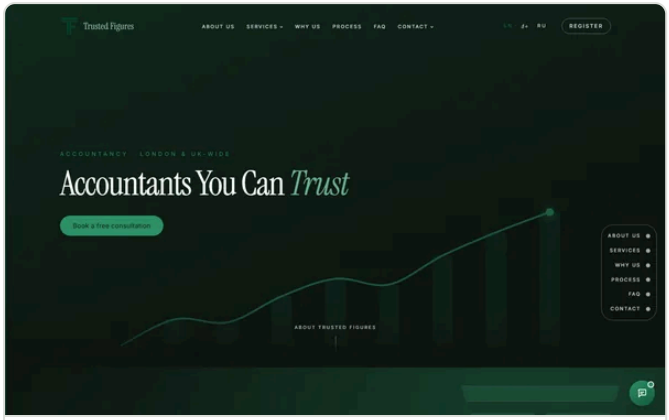
Decisions and trade-offs

- Template messages leave through two send workflows, not from each app. It adds a hop, but the secret check and the template allow-list sit in the send workflows instead of in every app.
- A failed WhatsApp reply cannot block a document from being saved: the two run separately. A client may wait for an answer, but no file is lost.
- The live workflows are exported to git, so changes can be reviewed outside n8n. The exports are only as current as the last export, but a broken edit can be compared with the last good copy.

Skills used in production

Python	01 CIS pipeline · 03 Bank statements
TypeScript	02 Chatbot
n8n	04 n8n automations · 02 Chatbot · 01 CIS pipeline
PostgreSQL / Supabase	01 CIS pipeline · 02 Chatbot · 04 n8n automations
LLM extraction with a rules cross-check	01 CIS pipeline
AI chatbots with guardrails and a fallback provider	02 Chatbot
LLM evaluation: acceptance benchmarks and scripted evaluation sets	02 Chatbot
Document reading: OCR and PDF text	01 CIS pipeline · 03 Bank statements
Financial data in exact decimals	01 CIS pipeline · 03 Bank statements
API integrations: WhatsApp Cloud API, Microsoft Graph, Companies House	02 Chatbot · 04 n8n automations
Browser tests with Playwright	01 CIS pipeline
CI and automated tests: GitHub Actions, pytest, strict type checks	01 CIS pipeline · 03 Bank statements · 02 Chatbot
AI-assisted development with Claude Code, with a second AI reviewing each piece of work	01 CIS pipeline · 02 Chatbot · 03 Bank statements
Websites in Next.js and Tailwind, multilingual with next-intl	Websites

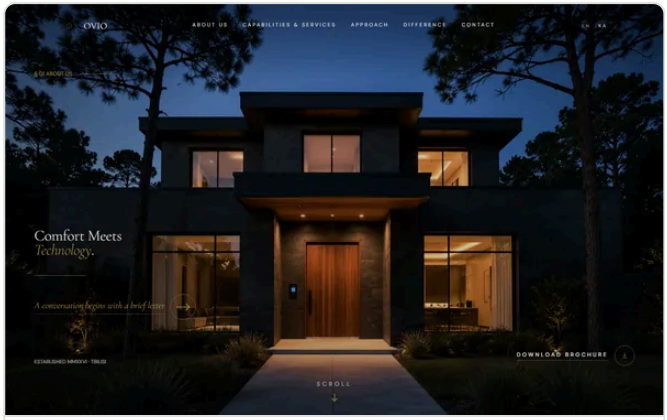
Websites



Trusted Figures

Multilingual site for a London accountancy firm, with online registration and document upload that feeds its automations.

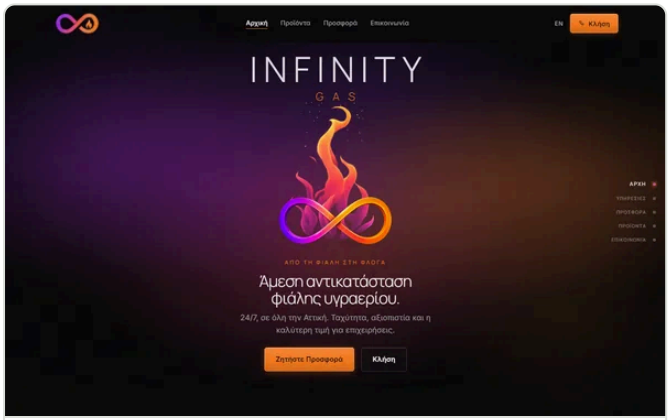
`Next.js, Supabase, Vercel`



OVIO Solutions

Bilingual Georgian and English site for a smart-home studio in Tbilisi, with a contact form that emails the studio.

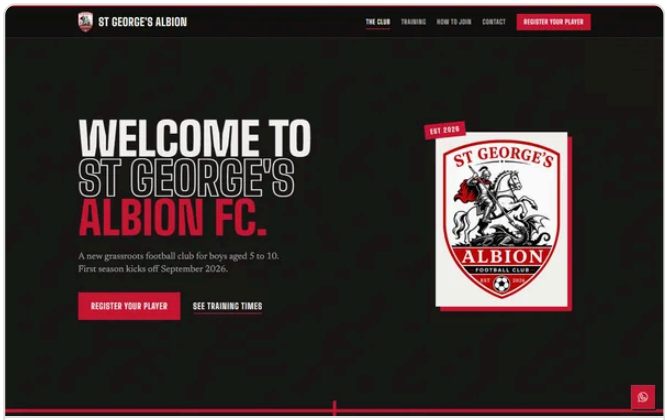
`Next.js, Tailwind, next-intl`



Infinity Gas

Greek and English site for a gas cylinder delivery business in Attica, built around a quote request form.

`Next.js, next-intl, Vitest`



St George's Albion FC

Site for a new youth football club where parents book a free session and register a player online.

`Next.js, Zod, Resend`

How I work

Every system starts from a written spec: what it does, what it must never do, and how we will know it works. A person signs off every figure before it is filed and every statement request before it is sent. Each system ships with handover documents, so someone else can run it.

I build with AI coding tools. Tests are written first and seen failing, and a second AI reviews each piece of work before it merges. What I own is the problem framing, the design and its safeguards, checking the results, and running the systems live.

Background

AI and Automation Specialist at Trusted Figures, a London accountancy firm. I started with simple scripts and n8n workflows in 2025. Production systems followed from February 2026, starting with a compliance tracker, and the WhatsApp assistant and the construction tax (CIS) statement pipeline from April 2026.

Languages: English, Georgian, Russian, Greek.

Contact

Write to me by email or on LinkedIn.

ladopaichadze@gmail.com / LinkedIn

How these figures were measured

1. 41: dashboards live on launch day, from the dated project timeline (at launch; measured Aug 2026)
2. 335 of 351: read-only count of documents received, by current outcome; final = matched, duplicate, replaced, not a CIS statement, annual statement or removed (some set by a person) (22 Jul to 13 Oct 2026, WhatsApp and email; measured Oct 2026)
3. 42 of 42: acceptance benchmark of 42 defined cases, run against the live AI model; cases passed (live AI model; measured Jul 2026)
4. 140 of 140: every statement's lines re-added to the bank's printed totals and running balance (14,801 transactions, 0 parse failures; measured Jul 2026)
5. 9: count of bank layouts the tool tries, and the brands each one names (11 UK bank brands, 10 tested on real statements; measured Aug 2026)
6. 14: workflows marked active in a read of the live instance (live n8n instance; measured Oct 2026)
7. 47: fixed set of real documents read again after each fix to the reader and compared with a person's recorded answers (real documents with a person's answers; measured Sep 2026)
8. 4,293: automated tests listed by the test runner on main (the whole CIS pipeline code; measured Oct 2026)
9. 62 of 64: read-only count of unique Q1 statements matched to an expected statement, over the Q1 statements identified; one match with its sender unconfirmed left out (Q1 2026-27, sender confirmed; measured Oct 2026)
10. 17: read-only count of documents held as duplicates: a re-sent file, or a second statement for the same month inside one PDF (since intake went live; measured Oct 2026)
11. under a minute: one timed run of the automated export, 3 clients at about 45 seconds each; before: about 4 minutes of manual download per company, Lado's estimate (per company, QuickBooks export; measured Oct 2026)
12. 6: count of languages in the assistant's supported list on the main branch (the WhatsApp assistant; measured Oct 2026)
13. 49: count of scripted client conversations in the evaluation set; 26 of them must pass on every run (the bot's real reply route on the live model; measured Oct 2026)
14. 1,689: automated tests counted in a local run with placeholder settings and no live services (the assistant and dashboard code; measured Oct 2026)
15. 8 of 8: photos stored and paired to their messages in a live test; two live tests before the fix stored 5 of 8 (live test after the fix; measured Sep 2026)
16. 976: automated tests listed by the test runner on main (the bank-statement processor's code; measured Oct 2026)
17. 14: active workflows grouped by job from a read of each one's steps: messages in 2, messages out 2, document filing 3, registrations 1, compliance 5, monitoring 1 (live n8n instance, by job; measured Oct 2026)